Gymnastiar Al Khoarizmy

Data Science Graduate
Institut Teknologi Sumatera

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GitHub Profile
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Summary

Data Science graduate from Institut Teknologi Sumatera with hands-on experience in machine learning, deep learning, MLOps, data processing, and model evaluation. Experienced in building applied AI projects across computer vision, speech understanding, time series forecasting, and big data analytics. Strong interest in AI data quality, data validation, labeling workflows, and contributing to production-oriented data science projects in a collaborative environment.

Education

- **Bachelor of Science in Data Science** — *2022 – 2026*
  Institut Teknologi Sumatera, Lampung — GPA: 3.81/4.00

Experience

- **Investment Data Analyst Intern** — *Jun 2025 – Sep 2025*
  Ministry of Investment and Downstream Industry / BKPM — Jakarta, Indonesia
  - Developed an Excel VBA-based monitoring tool to integrate and automate quarterly investment reporting compliance checks across company and project records.
  - Built a predictive machine learning model to estimate the transition time of investment projects from construction to production phase.
  - Processed investment datasets, performed feature engineering and model evaluation, and translated analytical results into insights for proactive project monitoring.
  - Presented analytical findings and model results to directors and mentors to support data-driven decision-making.
- **Data Science & Programming Practicum Assistant** — *Jul 2024 – Dec 2025*
  Institut Teknologi Sumatera — Lampung, Indonesia
  - Mentored students across programming, statistics, data mining, and data science-related practicum sessions, including Python, R, regression, classification, hypothesis testing, ANOVA, and model evaluation.
  - Assisted students in debugging code, preprocessing data, implementing statistical and machine learning models, and interpreting analytical results.
  - Evaluated weekly assignments and supported practical learning activities to improve students' technical understanding and problem-solving skills.

Projects

- **Zero-Shot Coastal Waste Detection and Material Classification** — *Final Project / Thesis*
  Grounding DINO, CLIP, Computer Vision, Model Evaluation
  - Developed a zero-shot AI pipeline to detect and classify coastal waste materials from 196 beach images in Lampung Province with 776 manually annotated objects across 6 material classes.
  - Evaluated prompt scenarios and model performance using IoU, precision, recall, F1-score, and classification accuracy.
  - Analyzed false positives, false negatives, class imbalance, and model behavior to assess data quality and improve interpretation of AI model results.
- **End-to-End Spoken Language Understanding for Retail Transaction Recording** — *Course Project*
  CNN-BiLSTM, MLOps, Deep Learning, Indonesian Speech Understanding — GitHub
  - Built an Indonesian spoken language understanding pipeline for retail transaction recording using deep learning-based sequence modeling.
  - Implemented an end-to-end workflow covering data preprocessing, model development, evaluation, and project documentation.
  - Applied MLOps concepts to support reproducibility, structured experimentation, and collaborative model development.
- **Plant Disease Diagnosis System using MLOps and Active Learning** — *INNOVEST 2026 – 3rd Winner*
  Machine Learning Operations, Active Learning, Data Labeling Workflow
  - Developed a machine learning-based plant disease diagnosis concept supported by MLOps and active learning to improve model development and data labeling efficiency.
  - Designed an iterative workflow where uncertain predictions can be reviewed and used to improve future model performance.
- **Predictive Flood Analytics with Hadoop Ecosystem in Lampung** — *Course Project*
  Hadoop Ecosystem, Big Data Analytics, Predictive Modeling — GitHub
  - Built a predictive analytics workflow using the Hadoop ecosystem for flood-related data analysis in Lampung.
  - Processed and analyzed multi-source data to support disaster risk analytics and predictive modeling.
- **Seasonal Forecasting of Ferry Passenger Demand** — *Research Project*
  Time Series Forecasting, Transportation Analytics, Research Publication — Article
  - Conducted time series forecasting to analyze seasonal ferry passenger demand at Bakauheni Port, Indonesia.
  - Developed forecasting insights to support operational planning and transportation demand analysis.

Achievements

- **3rd Winner** – INNOVEST 2026 "Clash of Champions", Politeknik Negeri Jember — *2026*
  - Awarded 3rd Winner for the project "Implementation of a Plant Disease Diagnosis System with Machine Learning Operations (MLOps) and Active Learning."

Technical Skills

Programming: Python, R, SQL
Data Science: Feature Engineering, Machine Learning, Deep Learning, Model Evaluation, Time Series Forecasting, Data Quality & Labeling
MLOps & Tools: Git, GitHub, Docker, Streamlit, Hadoop, Excel VBA
Visualization: Tableau, Looker Studio
Soft Skills: Analytical Thinking, Documentation, Communication, Collaboration
Languages: Indonesian (Native), English

Certifications

•**Python Project for Data Science** – IBM Coursera
•**Python for Data Science, AI & Development** – IBM Coursera
•**Introduction to Financial Accounting** – University of Pennsylvania, Coursera